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ENGINEERING



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Sreea Institute of Medical And Technical Sciences
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NETWORK TRAFFIC AND CONGESTION CONTROL IN MODERN COMMUNICATION SYSTEM

CSA0716-COMPUTER NETWORKS

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UNDER THE GUIDENCE OF:

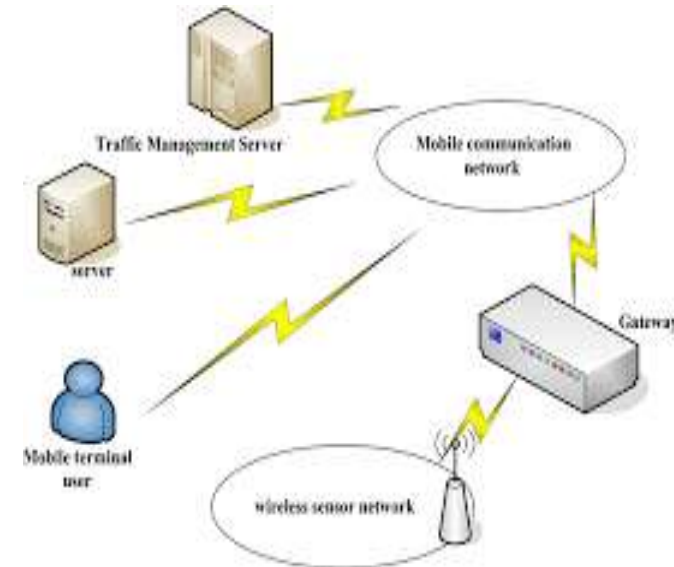
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INTRODUCTION



- Nowadays, many people use the internet for learning, work, and entertainment.
- Because of this a lot of data moves on the network every time.
- When too many users use the network together, congestion happens.
- Congestion makes the internet slow and causes delay and data loss.
- This project studies how to control traffic and improve network speed.





ABSTRACT



- This module focuses on how data is transmitted through networks and how traffic increases during heavy usage.
- The study discusses common causes of network congestion such as high data demand, limited bandwidth, and network faults.
- Different techniques and protocols used to control congestion and improve data flow are explained.
- The main aim is to know how proper traffic control helps in achieving better speed, reliability, and quality of service.



PROBLEM STATEMENT



- During peak hours, internet speed becomes very slow due to heavy user traffic.
- Online classes and video calls get disturbed because of network congestion.
- Packet loss occurs when too many users access the network at the same time.
- Mobile and Wi-Fi networks fail to provide stable connectivity in crowded areas.
- Banking and online payment systems face delays due to network overload.
- Smart devices and IoT systems stop working properly because of poor traffic management.



MODULE DESCRIPTION

❑ Module 1: Network Traffic Analysis

This module studies how data moves in the network and finds where traffic is high.

❑ Module 2: Network Congestion Management Testing

This module tests the network in different conditions to check delay, speed, and data loss.

❑ Module 3: Congestion Management Strategies

- This module uses some methods to control traffic and reduce congestion.
- These three modules work together to improve network performance.
- They help in providing fast, smooth, and reliable communication.



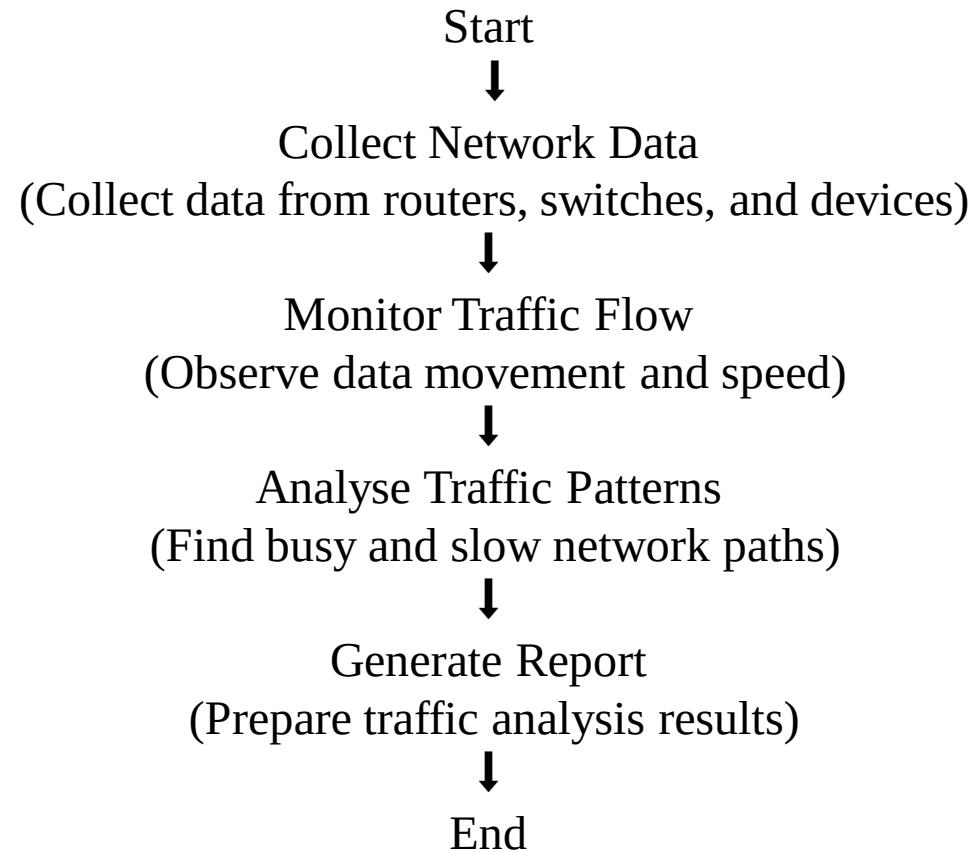
MODULE-1:NETWORK TRAFFIC ANALYSIS



- Network traffic analysis studies how data moves from one device to another.
- It helps in understanding the amount of data used in a network.
- It is used to find busy and slow parts of the network.
- It helps in improving network speed and performance.
- It supports better planning and management of communication systems.



MODULE 1: FLOW CHART

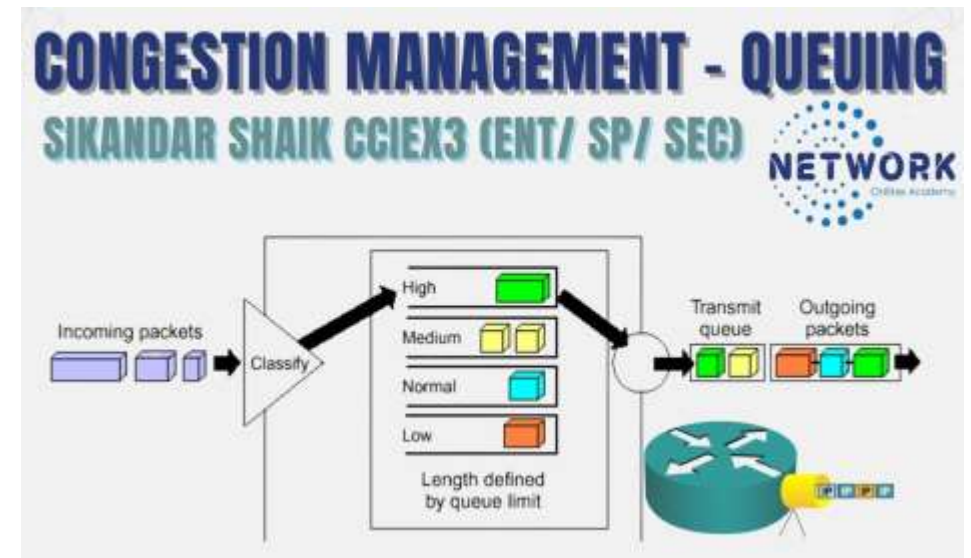




MODULE 2 : NETWORK CONGESTION MANAGEMENT

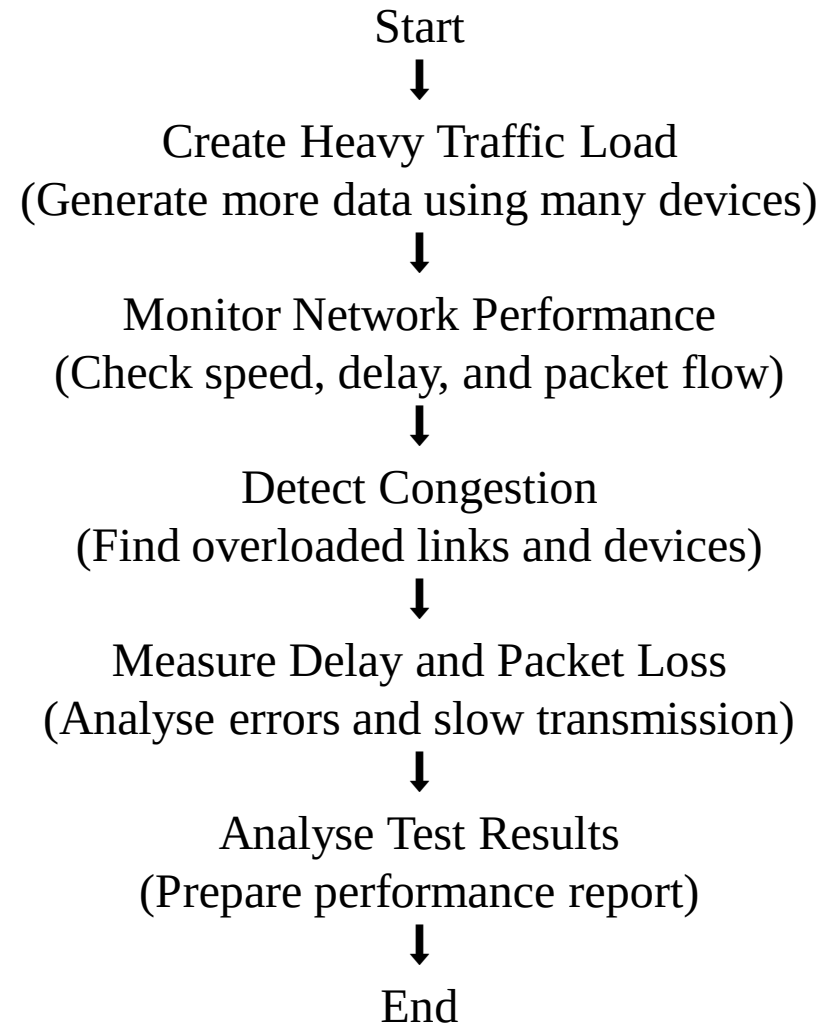


- Network congestion management testing checks how a network works under heavy traffic.
- It helps in finding when and where congestion occurs.
- It is used to measure delay, packet loss, and speed.
- It helps in improving network stability and performance.
- It ensures smooth data communication during peak usage.





MODULE 2 :FLOW CHART





MODULE 3 : CONGESTION MANAGEMENT STRATEGIES



- This module focuses on controlling heavy network traffic.
- It uses different techniques to help data move smoothly.
- It gives priority to important data packets.
- It prevents network overload and breakdown.
- It improves overall network performance.
- It balances traffic between different paths.



TECHNIQUES FOR CONGESTION MANAGEMENT



- Traffic Shaping
Controls the speed of data transmission to avoid sudden overload.
- Admission Control
Allows new users only when the network has enough capacity.
- Queue Management
Organizes packets in routers to reduce delay and packet loss.
- TCP Congestion Control
Adjusts sending speed based on network conditions.
- Active Queue Management (AQM)
Detects congestion early and controls traffic before overload.



RESULT



- Congestion control helps in reducing heavy traffic in the network, which improves overall data transmission.
- Proper management of packets reduces data loss during communication.
- By controlling traffic flow, delay in sending and receiving data is minimized.
- The network becomes more stable and does not fail easily during peak hours.
- Users experience better quality in services like video calls, online classes, and browsing.
- Network resources like bandwidth and routers are used more efficiently.



CONCLUSION



- This project explains how network traffic and congestion affect communication systems.
- This module shows the importance of analyzing traffic and testing network performance.
- Different strategies are used to control congestion and improve data flow.
- Proper congestion management reduces delay and packet loss.
- Effective traffic control ensures fast, stable, and reliable communication.



FUTURE SCOPE



- Advanced tools can be used for better network traffic monitoring.
- Artificial Intelligence can help in predicting and controlling congestion.
- High-speed networks like 5G and 6G will improve data transmission.
- More secure and reliable congestion control methods can be developed.
- Cloud-based systems can be used for efficient network management.





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